

Rethinking Wearability: When the Pillow Wears the Dreamer

Toward Intimate Interfaces and Passive Wearable Futures

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Abstract

This paper reimagines *wearability* by exploring how objects might *engage* the body through relational intimacy rather than attachment, and explores personalization not as a feature, but as an ontological function. We introduce PillowTalk 2.0, a sleep-based interface designed for dream recall as a design case. Unlike conventional wearables that assume an active, data-driven user, PillowTalk 2.0 engages passivity and stillness. We propose that objects may "wear" the user, advocating for a shift in wearable paradigms toward intuition, intimacy, and the unconscious body.

Ccs concepts

• Human-centered computing → Interaction design theory, concepts and paradigms

Keywords

Wearable technology; passive interfaces; personalization; dream recall; embodied interaction; intimate computing; speculative design; sleep interfaces

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1 Introduction

Wearability is often synonymous with control, tracking, or augmentation. Whether it is smartwatches, fitness bands, or biometric garments, wearable technologies typically assume an

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active user who monitors or manipulates their data in real time. But what happens when the most meaningful data emerges not through directed use, but through surrender and passivity? What if the wearable becomes not an accessory on the body, but an object attuned to moments when agency and control are suspended?

This paper reframes wearability as a question of relation. Pillows are seldom thought of as wearables, yet they meet many of the criteria: proximity, routine contact, somatic shaping. Deeply personal, they provide comfort, support, and familiarity, absorbing sweat, scent, and sleep patterns over time. Choosing a pillow is an exercise in individual preferences, considering factors such as texture, size, firmness, material, and shape. Safety, contentment, coziness, rest—we create attachments to our pillows. Historically, pillows have carried cultural, spiritual, and emotional significance, serving as more than mere comfort objects, but as vessels of memory and meaning, and as seats that hold our heads above the ground when at our most vulnerable. We suggest that pillows can be seen as *prosthetics of rest*, and in doing so, they offer a compelling framework for rethinking wearables beyond visibility or control.

With PillowTalk 2.0, we explore wearables that align not with activity, but with *receptivity*. In doing so, we challenge normative design logics and propose a more relational and intimate model of bodily interface.

1.1 PillowTalk 2.0

Remembering dreams allows access to a unique form of thought that is deeply connected to our waking life, offering valuable psychological meaning and potential for growth, healing, and creativity.

PillowTalk 2.0 is a dream-recording device designed to intuitively capture dreams during hypnopompia, the fleeting transition between sleep and wakefulness when dream imagery is most vivid. PillowTalk 2.0 focuses on low-effort, secure, and natural in-home dream capture, distinguishing itself from regular pillows and encouraging personal interaction through design and customization. Personalization acts as a cognitive primer, increasing users' ability to recall dreams [1, 2].

2 The Boundaries of Wearability

Wearables are typically defined as objects worn on the body that enhance, extend, or sense human capacities [3]. Smartwatches, fitness trackers, and AR glasses are becoming an integral part of our daily lives, embedded into our bodily rhythms. However, this definition implies a unidirectional relationship, where the human body acts as a platform and the object serves as an augmentation or instrument.

What happens when we reverse the direction of this relationship? Can an object "wear" the body, not just metaphorically, but materially and functionally?



Figure 1: Albrecht Dürer's Pillow Studies (1493)

1.1 The Pillow Wears the Head

Albrecht Dürer's *Pillow Studies* (1493) are a series of remarkably detailed renderings of crumpled, collapsing pillows, where the human figure is visibly implied but not depicted. The pillow, serving as a soft printmaker's plate that captures a negative presence, challenges the notion that the body is the active agent and the object is the passive recipient. The pillow reacts, reshapes, and remembers the body, mirroring how wearables capture biometrics, including heartbeats, motion, and temperature. However, here, the signals are static and sculptural.

If wearables are defined by proximity, continuity, and interaction, the pillow fits the definition as an object of intimacy, proximity, and recurrence. The pillow molds itself to our shape, absorbs body heat and scent, captures hair, sweat, and skin cells, and influences posture and breathing. Unlike most wearables, which are strapped on, the pillow holds and nests. To say "*the pillow wears the head*" is to invert the usual framing of agency. Who shapes whom? What is the site of interaction? Body, object, or space in between? Here, wearability becomes relational, not about which object is on which body, but about how object and

body co-create experience and how, in this enmeshment of embodied, ritualized use, personalization emerges.

1.2 Redefining Wearable Interfaces

Reframing the pillow as a wearable, or as a body-wearer, invites a new vocabulary for interaction design. It opens the door to "passive wearables," "ambient wearables," or "restful interfaces." It also destabilizes the boundaries of the body, suggesting that intimacy, duration, and co-adaptation may be just as important as functionality in defining wearable technology.

2 Personalization as Function: The Pillow as a Medium for Dream Recall

While much wearable design is optimized for one-size-fits-all models that ensure scalability, this paper proposes that personalization is not ornamental, but ontological to the function of certain objects.

PillowTalk 2.0's functionality rests not only in its embedded technology but in its ability to become a differentiated, intimate, and meaningful artifact. PillowTalk 2.0, then, is not a pillow, but a kit that can be embedded into any pillow. As such, we want to preserve the user's intimate relationship with their pillow, as well as inspire them to reimagine what a pillow is and can be, and to create and design their own.

2.1 Evocative Objects

Pillows and headrests are a site of meaningful symbolism, ritual significance, cultural expression and bodily enmeshment. Historically, their design, materials, and ornamentation were intricately tailored to individual and societal values, reflecting class distinctions, spiritual beliefs, and localized climates.

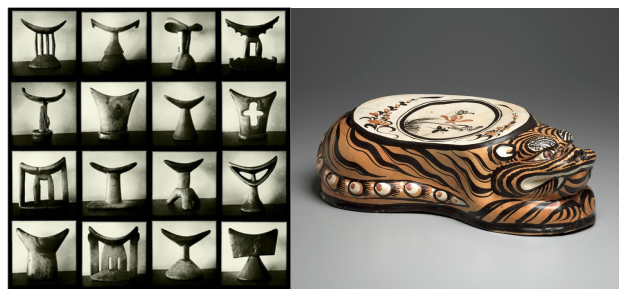


Figure 2: Collection of African Headrests and Tiger Pillow, Song Dynasty, late 12th Century, China.

Across historical contexts, from the ceramic headrests of Song Dynasty China to the ritualized wooden supports of ancient Egypt, pillows have not only accommodated the physical needs of comfort, cooling, and spinal support but have also protected hairstyles and warded off pests. Holding the head, seat of consciousness and identity, they have conveyed prayers, virtues, and aesthetic ideals. In China, ceramic pillows usually had popular songs, Confucian sayings, or poems inscribed on them. Meanwhile, children's pillows were made of cotton or silk stuffed

with grain husks and frequently took the form of a tiger, to protect the child from demons that cause nightmares [4, 5].

Today, choosing a pillow is an exercise in individual preferences, considering factors such as texture, size, firmness, material, and shape. High-end hotels, for example, offer pillow menus to accommodate the preferences of their guests, aware that a one-size-fits-all approach to certain objects will detract from their exclusivity.

In this broader context, the pillow is more than a passive object; it is a culturally embedded and meaning-laden interface between the body, the self, and the surrounding world.

2.2 Personalization as Priming

PillowTalk's primary function is to help users remember their dreams. One proven technique for enhancing dream recall is to remain physically still upon waking, which helps consolidate the dream memory. In addition, priming, that is, using specific cues before or during sleep, has been shown to significantly increase the likelihood of recalling dreams [6]. Even if the embedded technology fails, PillowTalk 2.0 can still function as a practical cognitive primer through its unique, personalized design. This differentiation alone can cue the brain toward dream-related associations.

2.3 Ergonomic Intimacy and Variability

Pillows, even the simplest of them, are inherently personalized. Unlike wristbands or glasses, pillows are not designed for public visibility, but for intimate ergonomics. Their form, size, fill, and firmness are widely varied because they interface with sensitive and variable behaviors, such as sleep position, breathing, neck curvature, and vulnerability. Chosen by feel and not by look, pillows extend the function of traditional wearables by catering to passive but continuous interaction. PillowTalk combines ergonomics and aesthetics to enhance each other.

2.4 Material Culture and Ritual

Pillows have carried prayers, amulets, herbs, protective embroidery, and ritual function for millennia. In many cultures, people place crystals, flower sachets, or tokens under their pillows, to alter sleep, invite visions, protect from spirits, or incubate dreams. PillowTalk 2.0 draws on this lineage: personalization as invocation, not of deities, but of self-awareness and memory.

2.5 Interfacing with Hypnompia

Designing an object intended to function just after waking up but before becoming fully alert presents its challenges. At the most basic, the user interface needs to be extremely simple and intuitive, without complex navigation or multi-step processes. Controls should be easily recognizable and accessible, with minimal cognitive demands of decision-making or attention.

While sleeping, users may accidentally activate features; therefore, the design should include safeguards to prevent errors and provide easy ways to recover from them, ideally without requiring a full wake-up.

Moreover, the act of *speaking* a dream poses questions of privacy, control, and vulnerability, particularly for users who do not sleep alone. Personalization, then, isn't just about colors or form. It's also about agency. The choice of activation modality offers users different thresholds of exposure. A voice-activated pillow might feel intrusive to one person and liberating to another. By allowing different gestural activations, the system respects bodily autonomy and context.

Conclusion

PillowTalk 2.0 invites us to extend our notion of wearability beyond something affixed to the body, to something that coexists with it intimately, receptively, and unconsciously. It does so by shifting the focus from active monitoring to passive priming, and from utility to relation. Pillows, long associated with rest, ritual, and personal meaning, offer an ideal model for an interface design grounded in familiarity, material culture, and somatic memory.

PillowTalk 2.0 reframes personalization as a condition of function rather than a consumer-facing feature by extending cognitive access to our inner worlds. As we blur the boundaries between user and object, interaction and ritual, technology and intimacy, we propose a more nuanced understanding of "wearables" where to wear something is also to be worn by it, and where the quietest interactions may carry the most profound meaning.

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